Gamma Distribution

- Continuous distribution
- **R.Vr. is always positive (defined for X in $(0,\infty)$)**
- **Defined as Skewed distribution**
- **Occurs as waiting time between events**
- **Also predict the wait time until future events occur**
- **Predicts the wait time until the k^{th} event occurs**

A **skewed distribution** is a type of probability distribution where the data points are not symmetrically distributed around the mean. It refers to the asymmetry in the shape of the distribution.

1. **Continuous Distribution**: The Gamma distribution is a continuous probability distribution defined for positive real values, typically representing wait times.
2. **Positive Random Variable**: The random variable X in the Gamma distribution is always positive, covering the range $X \in (0, \infty)$.
3. **Skewed Distribution**: It is generally right-skewed, with the skewness decreasing as the shape parameter k increases, approaching normality for large k.
4. **Waiting Time Between Events**: The Gamma distribution is often used to model the time between events in a Poisson process, making it suitable for wait-time predictions.
5. **Future Event Wait Time**: It can predict how long you will wait for future events in processes where events happen at a constant rate.
6. **Wait Time for kth Event**: The Gamma distribution is used to model the waiting time until the k th event occurs in a Poisson process, where k is the shape parameter.

Some properties of gamma function and its properties

The **gamma function** is defined by

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx, \qquad \text{for } \alpha > 0.$$

(a) $\Gamma(n) = (n-1)(n-2)\cdots(1)\Gamma(1)$, for a positive integer n.

(b) $\Gamma(n) = (n-1)!$ for a positive integer n.

(c) $\Gamma(1) = 1$. (d) $\Gamma(1/2) = \sqrt{\pi}$.

$$\Gamma(n+1) = n \cdot \Gamma(n)$$

Find the value of $\Gamma\left(\frac{3}{2}\right)$ using the known properties of the Gamma function.

Gamma Distribution

Gamma Distribution The continuous random variable X has a **gamma distribution**, with parameters α and β, if its density function is given by

$$f(x;\alpha,\beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, & x > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

where $\alpha > 0$ and $\beta > 0$.

- **Alpha (α)**: Shape parameter that affects the form of the distribution.
- **Beta (β)**: Scale parameter that stretches or compresses the distribution.

Gamma(α, β) or **Gamma(k, β)** or **Γ(k, β)**

Note: Graphs of several gamma distributions are shown in Figure for certain specified values of the parameters α and β. The special gamma distribution for which $\alpha = 1$ is called the **exponential distribution**.

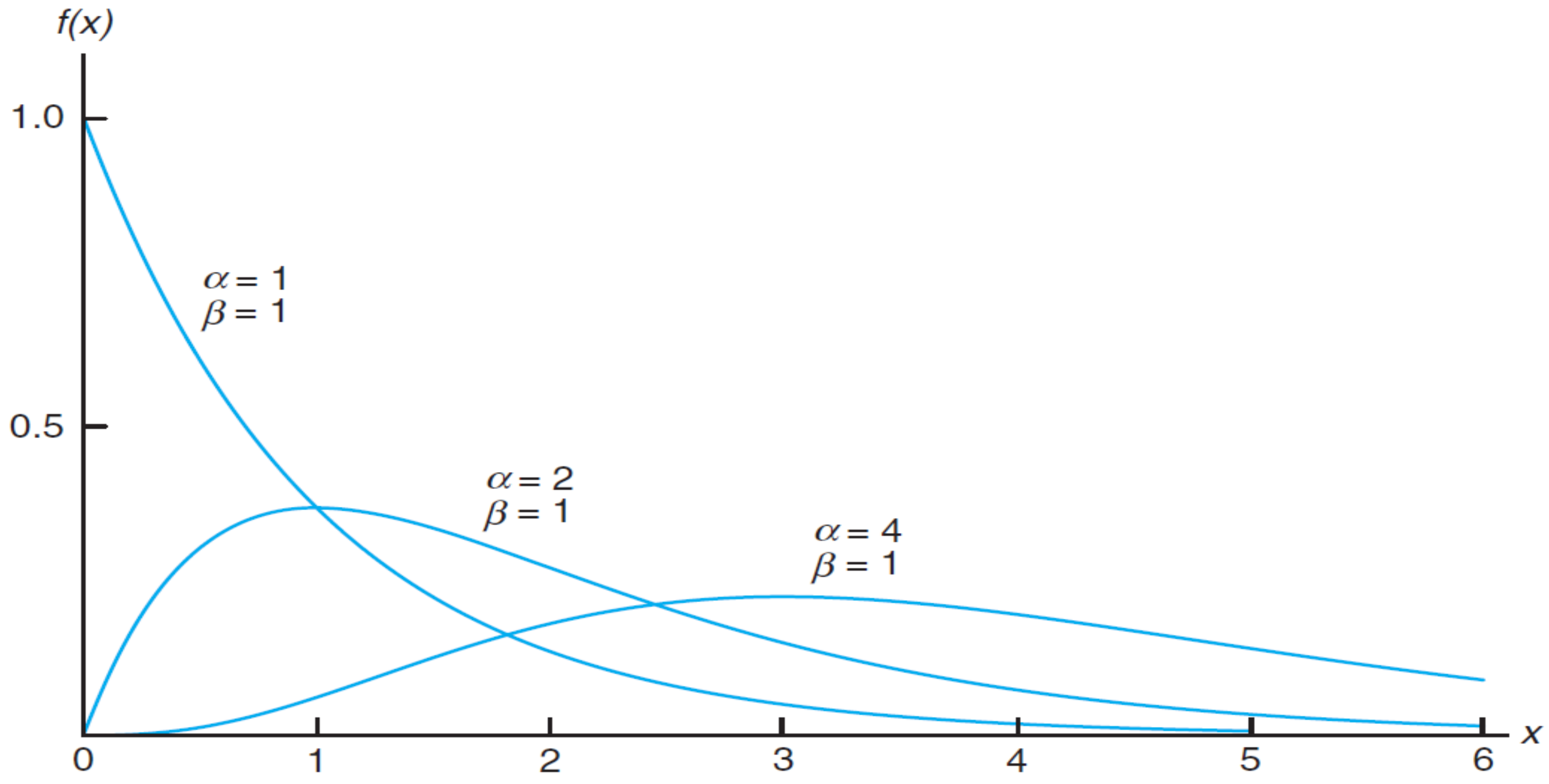

Figure 6.28: Gamma distributions.

Exponential Distribution

The continuous random variable X has an **exponential distribution**, with parameter β, if its density function is given by

$$f(x;\beta) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & x > 0, \\ 0, & \text{elsewhere}, \end{cases}$$

where $\beta > 0$.

Mean, Variance and MGF:

The mean and variance of the gamma distribution are

$$\mu = \alpha\beta \text{ and } \sigma^2 = \alpha\beta^2.$$

The mean and variance of the exponential distribution are

$$\mu = \beta \text{ and } \sigma^2 = \beta^2.$$

The moment generating function of gamma distribution is $M_X(t) = (1 - t\beta)^{-\alpha}$

The moment generating function of exponential distribution is $M_X(t) = (1 - t\beta)^{-1}$

2. Exponential Distribution

Definition: A continuous RV X is said to follow an *exponential distribution* or *negative exponential distribution* with parameter $\lambda > 0$, if its probability density function is given by

$$f(x) = \begin{cases} \lambda\, e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

We note that $\int_0^\infty f(x)\, dx = \int_0^\infty \lambda\, e^{-\lambda x}\, dx = 1$ and hence $f(x)$ is a legitimate density function.

The mean of the exponential distribution is $1/\lambda$

The variance of the exponential distribution is $1/\lambda^2$

Self study-

Relationship to the Poisson Process

We shall pursue applications of the exponential distribution and then return to the gamma distribution. The most important applications of the exponential distribution are situations where the Poisson process applies (see Section 5.5). The reader should recall that the Poisson process allows for the use of the discrete distribution called the Poisson distribution. Recall that the Poisson distribution is used to compute the probability of specific numbers of "events" during a particular *period of time or span of space*. In many applications, the time period or span of space is the random variable. For example, an industrial engineer may be interested in modeling the time T between arrivals at a congested intersection during rush hour in a large city. An arrival represents the Poisson event.

Applications of the Exponential and Gamma Distributions

In the foregoing, we provided the foundation for the application of the exponential distribution in "time to arrival" or time to Poisson event problems. We will illustrate some applications here and then proceed to discuss the role of the gamma distribution in these modeling applications. Notice that the mean of the exponential distribution is the parameter β, the reciprocal of the parameter in the Poisson distribution. The reader should recall that it is often said that the Poisson distribution has no memory, implying that occurrences in successive time periods are independent. The important parameter β is the mean time between events. In reliability theory, where equipment failure often conforms to this Poisson process, β is called **mean time between failures**. Many equipment breakdowns do follow the Poisson process, and thus the exponential distribution does apply. Other applications include survival times in biomedical experiments and computer response time.

Book: Probability & Statistics for Engineers & Scientists by Raymond Myers and Ronald E. Walpole

The **Gamma distribution** models the waiting time until the $\boldsymbol{k}^{\text{th}}$ event occurs in a **Poisson process** with rate $\boldsymbol{\lambda}$. Specifically, suppose the time between events follows an Exponential distribution with parameter λ. In that case, the sum of $\boldsymbol{k}$ independent exponentially distributed random variables follows a **Gamma distribution** with shape parameter $\boldsymbol{k}$ and scale parameter $\beta=1/\lambda$. Additionally, the Gamma distribution serves as a conjugate prior for the Poisson distribution in Bayesian inference.

Remarks-

- The **time** between **events** in a **Poisson process** is an exponential distribution
- Poisson- Event per single unit of time
- Exponential – Time per single event
- The time required for the first event to occur is an exponential distribution
- The **mean** of an **exponential distribution** is a parameter **beta**, the **reciprocal** of the **parameter** in the **Poisson** distribution
- Beta is called the mean time between events
- The time until the **occurrence** of a **Poisson event** is a random variable described by **exponential** distribution whereas the **time** until a specified number of **Poisson events** occur is a random variable described by a **gamma** distribution
- The **specific number** of **events** is the parameter **alpha** in the gamma distribution

Remarks-

- **Poisson Distribution**: Models the number of events occurring in a fixed unit of time or space.
- **Exponential Distribution**: Describes the time between consecutive events in a Poisson process; specifically, it models the time until the first event.
- **Relationship Between Poisson and Exponential**: The time between events in a Poisson process follows an Exponential distribution, with the mean being the reciprocal of the Poisson event rate ($\beta = \frac{1}{\lambda}$).
- **Beta in Exponential Distribution**: Represents the mean time between events in an Exponential distribution, which is the reciprocal of the rate in the Poisson distribution.
- **Gamma Distribution**: Describes the time until a specified number (α) of Poisson events occur, generalizing the Exponential distribution for multiple events.
- **Alpha in Gamma Distribution**: Represents the number of Poisson events until which the wait time is modeled.

The mileage which car owners get with a certain kind of radial tire is a RV having an exponential distribution with mean 40,000 km. Find the probabilities that one of these tires will last (a) at least 20,000 km, and (b) at most 30,000 km.

Let X denote the mileage obtained with the tire

$$f(x) = \frac{1}{40,000} e^{-x/40,000} \quad x > 0$$

(a) $$P(X \geq 20,000) = \int_{20,000}^{\infty} \frac{1}{40,000} e^{-x/40,000}\, dx$$

$$= [-e^{-x/40,000}]_{20,000}^{\infty}$$

$$= e^{-0.5} = 0.6065$$

(b) $$P(X \leq 30,000) = \int_{0}^{30,000} \frac{1}{40,000} e^{-x/40,000}\, dx$$

$$= [-e^{-x/40,000}]_{0}^{30,000}$$

$$= 1 - e^{-0.5} = 0.5270$$

Example 1

In a certain city, consumption of electric power, in millions of kilowatt-hours, is a random variable X having a Gamma distribution with a mean μ=6 and variance σ^2=12. Find the values of α and β

Example 2

Suppose that telephone calls arriving at a particular switchboard follow a Poisson process with an average of 5 calls coming per minute. What is the probability that up to a minute will elapse by the time 2 calls have come into the switchboard?

Example 3

6.41 If a random variable X has the gamma distribution with $\alpha = 2$ and $\beta = 1$, find $P(1.8 < X < 2.4)$.

6.42 Suppose that the time, in hours, required to
repair a heat pump is a random variable X having
a gamma distribution with parameters $\alpha = 2$ and
$\beta = 1/2$. What is the probability that on the next service call
(a) at most 1 hour will be required to repair the heat pump?
(b) at least 2 hours will be required to repair the heat pump?

PRACTICE QUESTIONS

In a biomedical study with rats, a dose-response investigation is used to determine the effect of the dose of a toxicant on their survival time. The toxicant is one that is frequently discharged into the atmosphere from jet fuel. For a certain dose of the toxicant, the study determines that the survival time, in weeks, has a gamma distribution with $\alpha = 5$ and $\beta = 10$. What is the probability that a rat survives no longer than 60 weeks?

6.81 The length of time between breakdowns of an essential piece of equipment is important in the decision of the use of auxiliary equipment. An engineer thinks that the best model for time between breakdowns of a generator is the exponential distribution with a mean of 15 days.

(a) If the generator has just broken down, what is the probability that it will break down in the next 21 days?

(b) What is the probability that the generator will operate for 30 days without a breakdown?

A light bulb manufacturer claims that their bulbs have an average lifetime of 800 hours. Calculate the probability that a light bulb lasts longer than 1000 hours.

6.45 The length of time for one individual to be served at a cafeteria is a random variable having an exponential distribution with a mean of 4 minutes. What is the probability that a person is served in less than 3 minutes on at least 4 of the next 6 days?

Suppose that a study of a certain computer system reveals that the response time, in seconds, has an exponential distribution with a mean of 3 seconds.

(a) What is the probability that response time exceeds 5 seconds?

(b) What is the probability that response time exceeds 10 seconds?

The lifetime of a certain type of battery follows an Exponential distribution with a mean of 500 hours. What is the probability that a battery lasts less than 400 hours?

A machine has a failure rate of 0.2 failures per hour. What is the probability that the machine will operate without a failure for at least 5 hours?

Suppose that a study of a certain computer system reveals that the response time, in seconds, has an exponential distribution with a mean of 3 seconds.

MCQs on Gamma Distributions

1. The value of $\Gamma(\frac{3}{2})$=

a) $\frac{\sqrt{\pi}}{2}$ b) $\frac{\sqrt{\pi}}{3}$ c) $\frac{3\sqrt{\pi}}{2}$ d) none

2. If mean and variance of the gamma distribution is given as 24 and 192 respectively. Then the values of parameters α (shape) and β (scale) are

a) $\alpha = 8$, $\beta = 3$

b) $\alpha = 3$, $\beta = 8$

c) $\alpha = 9$, $\beta = 8$

d) *none*

3. If random variable X has a Gamma distribution with $\alpha = 2$ and $\beta = 3$. Then the mean and standard deviation $(S.D)$ of X are

a) mean= 6 , S.D.= 18

b) mean= 18 , S.D.= 6

c) mean= 6 , S.D.= 4.2426

d) mean=12, S.D.=4.2426

4. On Sunday morning, customers arrive at cafeteria follows a Poisson process with average rate 15 customers per hour. What is the average amount of time elapse before 3 customers arrive in cafeteria.

a) 4 mins b) 9 mins c) 12 mins d) 16mis

MCQs on Exponential distributions

1. **If a continuous random variable X has an exponential distribution with mean 0.5. Then P(X>1/2)=**

a) e^{-1} b) $e^{-1/2}$ c) (1/2) e^{-1} d) [1/2] $e^{-1/2}$

2. **Suppose that a system contains a certain type of component whose lifetime is T years. The random variable T is modelled nicely by the exponential distribution with mean time to failure is $\beta = 5$.**
The probability that a given component installed on a system is still working after 8 years is

a) 0.16 b) 0.17 c) 0.18 d) 0.2

3. **A continuous random variable X has probability density function given by**

$$f(x) = \begin{cases} 2e^{-2x}, & x \geq 0 \\ 0, & otherwise \end{cases}$$

The mean and variance of X are

a) ½ , 1/8 b) ½ , ¼ c) 1, ½ d) 2, ½

3. **The lifetime T of an alkaline battery is exponentially distributed with θ=0.05 per hour. What are mean and standard deviation of batteries lifetime?**

a) mean=10, S.D.=20 c) mean=20, S.D.=10
b) mean=20, S.D.=20 d) mean=10, S.D.=10